

Q1

Determine the time complexity:

```
for(int i=0;i<n;i++)  
{  
    cout<<i;  
}
```

Ans Single loop, 0 to n $O(n)$

Q2

Determine the time complexity:

```
for(int i=0;i<n;i++)  
{  
    for(int j=0;j<n;j++)  
    {  
        cout<<"*";  
    }  
}
```

Ans Nested loop, both 0 to n $O(n^2)$

Q3

Determine the time complexity:

```
for(int i=0;i<n;i++)  
{  
    for(int j=0;j<i;j++)
```

```
{  
    cout<<"*";  
}  
}
```

Ans Single loop, i multiplies by 2 $O(n^2)$

Q4

Determine the time complexity:

```
for(int i=1;i<n;i*=2)  
{  
    cout<<i;  
}
```

Ans $\Theta(\log n)$

Q5

Determine the time complexity:

```
for(int i=n;i>1;i/=2)  
{  
    cout<<i;  
}
```

Ans $\Theta(\log n)$

Q6

Determine the time complexity:

```
for(int i=1;i<=n;i++)  
{  
    for(int j=1;j<=log(n);j++)  
    {  
        cout<<"*";  
    }  
}
```

Ans $O(n \log n)$

Q7

Determine the time complexity:

```
while(n>0)  
{  
    n--;  
}
```

Ans $\Theta(n)$

Q8

Determine the time complexity:

```
while(n>1)  
{  
    n/=2;  
}
```

Ans $O(\log n)$

Q9

Determine the time complexity:

```
for(int i=1;i<=n;i++)  
{  
}  
for(int j=1;j<=n;j++)  
{  
}
```

Ans $O(n)$

Q10

Determine the time complexity:

```
for(int i=1;i<=n;i++)  
{  
}  
for(int j=1;j<=n*n;j++)  
{  
}
```

Ans $O(n^2)$

Q11

Determine the time complexity:

```
for(int i=1;i<=n;i++)  
{  
    for(int j=1;j<=n*n;j++)
```

```
{  
}  
}
```

Ans $O(n^3)$

Q12

Determine the time complexity:

```
for(int i=1;i<=n;i*=3)  
{  
}
```

Ans $O(\log n)$

Q13

Determine the time complexity:

```
for(int i=1;i<=n;i++)  
{  
    for(int j=1;j<=i*i;j++)  
    {  
    }  
}
```

Ans $O(n^3)$

Q14

Determine the time complexity:

```
void fun(int n)
```

```
{  
    if(n==0)  
        return;  
  
    fun(n-1);  
}
```

Ans $O(n)$

Q15

Determine the time complexity:

```
void fun(int n)  
{  
    if(n<=1)  
        return;  
  
    fun(n/2);  
}
```

Ans $O(\log n)$

Q16

Determine the time complexity:

```
void fun(int n)  
{  
    if(n==0)  
        return;
```

```
    fun(n-1);  
    fun(n-1);  
}
```

Ans $O(2^n)$

Q17

Determine the time complexity:

```
int fib(int n)  
{  
    if(n<=1)  
        return n;  
  
    return fib(n-1)+fib(n-2);  
}
```

Ans $O(2^n)$

Q18

Determine the time complexity:

```
int fact(int n)  
{  
    if(n<=1)  
        return 1;  
  
    return n*fact(n-1);  
}
```

Ans $O(n)$

Q19

Determine the time complexity:

```
void fun(int n)
{
    if(n<=0)
        return;

    fun(n-1);

    cout<<n;
}
```

Ans $O(n)$

Q20

Determine the time complexity:

```
void fun(int n)
{
    if(n<=1)
        return;

    fun(n/2);

    cout<<n;
}
```

Ans $O(\log n)$

Q21

Solve the recurrence relation:

$$T(n) = T(n-1) + 1$$

Ans $O(n)$

Q22

Solve the recurrence relation:

$$T(n) = T(n-1) + n$$

Ans $O(n^2)$

Q23

Solve the recurrence relation:

$$T(n) = T(n/2) + 1$$

Ans $O(\log n)$

Q24

Solve the recurrence relation:

$$T(n) = 2T(n-1) + 1$$

$O(2^n)$

Q25

Solve the recurrence relation:

$T(n) = T(n/2) + n$

Ans $O(n)$

Bonus GATE-Level Questions

Q26

```
for(int i=1;i<=n;i++)
{
    for(int j=1;j<=i;j*=2)
    {
    }
}
```

Ans

Outer 1 to n, inner $j*=2$ while $j \leq i$

$O(n \log n)$

Determine the complexity.

Q27

```
for(int i=1;i<=n;i*=2)
{
    for(int j=1;j<=n;j++)
    {
    }
}
```

Ans

Outer $i*=2$ while $i \leq n$, inner 1 to n

$O(n \log n)$

Determine the complexity.

Q28

```
for(int i=n;i>=1;i--)  
{  
    for(int j=i;j>=1;j--)  
    {  
    }  
}
```

Ans Outer n down to 1, inner i down to 1 $O(n^2)$

Determine the complexity.

Q29

```
for(int i=1;i<=n;i++)  
{  
    for(int j=1;j<=n;j*=2)  
    {  
    }  
}
```

Ans Outer 1 to n, inner $j*=2$ while $j \leq n$ $O(n \log n)$

Determine the complexity.

Q30

```
void fun(int n)  
{  
    if(n<=1)  
        return;
```

```
fun(n/2);  
fun(n/2);  
}
```

Ans Two recursive calls, each $n/2$ $O(n)$